# Research Evidence Pack

NotebookLM-compatible PDF generated from all worksheets in the research workbook

Purpose: preserve the structured evidence, numerical results, literature matrix, search log, practical requirements, and action plan in a file format that can be uploaded to NotebookLM.

## Evidence handling rules

• v25, v26, and earlier Seed 1/2 technical-audit results are kept separate. • Values are transcribed from the workbook without recalculation. • The research-gap matrix is a targeted audit, not a completed systematic review. • Preliminary findings are not evidence of general algorithmic superiority. • Verify important claims against the original papers, code, and logs.

Source workbook: Research_Results.xlsx

# Start Here

## Review scope

v29 matrix: 21 entries (17 targeted additions + 4 carry-forward anchors).

Not a systematic review; native database exports and full screening counts not complete.

## Current code package

v26 iFogSim2; 192 Python tests reported passing; 448 Java source files compiled; two pre-existing deprecation warnings.

The report uses supplied logs/README; no full rerun performed during synthesis.

## Training/evaluation

3 training seeds 801–803; 81-state Q table; v26 60 runs (2 scenarios × 6 named policies × 5 test seeds).

deadline_greedy duplicates latency_greedy; effectively five unique policies.

## Older experiment set

v25 summary: 4 scenarios × 5 policy mechanisms × 5 seeds = 100 run combinations.

Separate from v26; do not pool versions without a protocol/version audit.

## Legacy technical audit

Seed 1/2; 228 generated tasks per run; cutoff 500; old fixed-deadline setting.

History only; not the same experiment as v26 heterogeneous deadlines.

## Policy naming

gamma=0 and decision_model=contextual_bandit in current v26 config.

Call it tabular contextual-bandit/Q-value policy unless sequential Q-learning is reimplemented with tested gamma>0.

## Main performance result

Q policy wins mean latency on v25 edge capacity ×0.5; loses to latency-greedy on base/higher load and both new v26 scenarios; equal to greedy on network ×2.

No claim of general superiority.

## Research gap

No broad algorithmic gap confirmed in the current focused literature audit.

Supervisor must choose new algorithm vs comparative/reproducibility study.

## Metrics caveats

Latency ends at first processing-module completion; utilization and traffic are proxies; energy not task-level logged.

No full application-loop, CPU-utilization, total per-hop traffic, or energy-benefit claims yet.

## Next action

Freeze one version/config/model, close literature evidence gaps, agree metric definitions, run final independent seeds/topologies.

Do not tune or choose the method based on final test seeds.

# v25 Results

Archived v25 results — keep separate from v26

## Base / baseline workload | Tabular Q-value / contextual-bandit policy

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6491 |
| **Mean latency** | 10.662934693995672 |
| **Mean CI95 low** | 10.581399887455351 |
| **Mean CI95 high** | 10.744469500535994 |
| **P95 latency** | 24.38025903829714 |
| **P99 latency** | 36.84858958020693 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Base / baseline workload | Latency-greedy heuristic

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6491 |
| **Mean latency** | 8.274909413755847 |
| **Mean CI95 low** | 8.184800937706 |
| **Mean CI95 high** | 8.365017889805694 |
| **P95 latency** | 24.333333333333336 |
| **P99 latency** | 37.010892261324486 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Base / baseline workload | Edge-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6491 |
| **Mean latency** | 14.077350551585758 |
| **Mean CI95 low** | 13.953661027903976 |
| **Mean CI95 high** | 14.20104007526754 |
| **P95 latency** | 40.893434884787766 |
| **P99 latency** | 58.08012438937932 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Base / baseline workload | Local-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6491 |
| **Mean latency** | 20.314353527702465 |
| **Mean CI95 low** | 19.729298248253063 |
| **Mean CI95 high** | 20.899408807151858 |
| **P95 latency** | 98.28558198921826 |
| **P99 latency** | 123.07624883548928 |
| **Deadline violation rate** | 0.0596223885355212 |
| **Completion rate** | 1 |

## Base / baseline workload | Cloud-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6491 |
| **Mean latency** | 87.20124928689434 |
| **Mean CI95 low** | 86.80140367759678 |
| **Mean CI95 high** | 87.6010948961919 |
| **P95 latency** | 130.18788355720187 |
| **P99 latency** | 133.13155728169028 |
| **Deadline violation rate** | 0.4989993209174785 |
| **Completion rate** | 1 |

## Edge capacity 50% | Tabular Q-value / contextual-bandit policy

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6519 |
| **Mean latency** | 11.654437330078334 |
| **Mean CI95 low** | 11.453245222062291 |
| **Mean CI95 high** | 11.855629438094375 |
| **P95 latency** | 37.66666666666664 |
| **P99 latency** | 52.01903293778379 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Edge capacity 50% | Latency-greedy heuristic

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6519 |
| **Mean latency** | 13.994159292985636 |
| **Mean CI95 low** | 13.547566268720674 |
| **Mean CI95 high** | 14.4407523172506 |
| **P95 latency** | 37.66666666666665 |
| **P99 latency** | 102.70366716228838 |
| **Deadline violation rate** | 0.0190227129378348 |
| **Completion rate** | 1 |

## Edge capacity 50% | Edge-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6519 |
| **Mean latency** | 86.06076327287673 |
| **Mean CI95 low** | 73.65395109062783 |
| **Mean CI95 high** | 98.46757545512564 |
| **P95 latency** | 337.7980630777823 |
| **P99 latency** | 708.0901896512614 |
| **Deadline violation rate** | 0.2193266852725098 |
| **Completion rate** | 1 |

## Edge capacity 50% | Local-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6519 |
| **Mean latency** | 20.654021808111548 |
| **Mean CI95 low** | 19.847317062830047 |
| **Mean CI95 high** | 21.460726553393048 |
| **P95 latency** | 98.50595968067084 |
| **P99 latency** | 123.7785278878508 |
| **Deadline violation rate** | 0.0607452271999612 |
| **Completion rate** | 1 |

## Edge capacity 50% | Cloud-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6519 |
| **Mean latency** | 87.37002130042478 |
| **Mean CI95 low** | 86.91603920729699 |
| **Mean CI95 high** | 87.82400339355257 |
| **P95 latency** | 130.37127805040373 |
| **P99 latency** | 133.17886171227437 |
| **Deadline violation rate** | 0.5008576809369067 |
| **Completion rate** | 1 |

## Higher offered load (arrival interval ×0.7) | Tabular Q-value / contextual-bandit policy

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 9255 |
| **Mean latency** | 11.450311703360692 |
| **Mean CI95 low** | 11.307783828358568 |
| **Mean CI95 high** | 11.59283957836282 |
| **P95 latency** | 25.52542456694369 |
| **P99 latency** | 45.35441742269482 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Higher offered load (arrival interval ×0.7) | Latency-greedy heuristic

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 9255 |
| **Mean latency** | 9.552993781160602 |
| **Mean CI95 low** | 9.396136286944886 |
| **Mean CI95 high** | 9.709851275376318 |
| **P95 latency** | 24.39830714819479 |
| **P99 latency** | 39.58060026604058 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Higher offered load (arrival interval ×0.7) | Edge-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 9255 |
| **Mean latency** | 15.268535380052764 |
| **Mean CI95 low** | 15.022421314259711 |
| **Mean CI95 high** | 15.514649445845816 |
| **P95 latency** | 50.33528061233056 |
| **P99 latency** | 67.01050108529003 |
| **Deadline violation rate** | 0.0002158661629789 |
| **Completion rate** | 1 |

## Higher offered load (arrival interval ×0.7) | Local-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 9255 |
| **Mean latency** | 126.099601087379 |
| **Mean CI95 low** | 118.4779783598441 |
| **Mean CI95 high** | 133.72122381491388 |
| **P95 latency** | 513.1660294969108 |
| **P99 latency** | 1118.6062696057063 |
| **Deadline violation rate** | 0.2708728134415823 |
| **Completion rate** | 1 |

## Higher offered load (arrival interval ×0.7) | Cloud-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 9255 |
| **Mean latency** | 87.65298907558731 |
| **Mean CI95 low** | 87.39816395437983 |
| **Mean CI95 high** | 87.9078141967948 |
| **P95 latency** | 130.2661335966226 |
| **P99 latency** | 133.1099232239332 |
| **Deadline violation rate** | 0.5026506314117029 |
| **Completion rate** | 1 |

## Network propagation delay ×2 | Tabular Q-value / contextual-bandit policy

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6472 |
| **Mean latency** | 11.083174776092552 |
| **Mean CI95 low** | 10.872019570021454 |
| **Mean CI95 high** | 11.29432998216365 |
| **P95 latency** | 29.333333333333258 |
| **P99 latency** | 43.072558565505474 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Network propagation delay ×2 | Latency-greedy heuristic

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6472 |
| **Mean latency** | 11.083174776092552 |
| **Mean CI95 low** | 10.872019570021454 |
| **Mean CI95 high** | 11.29432998216365 |
| **P95 latency** | 29.333333333333258 |
| **P99 latency** | 43.072558565505474 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Network propagation delay ×2 | Edge-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6472 |
| **Mean latency** | 19.136677013625423 |
| **Mean CI95 low** | 19.005151296877496 |
| **Mean CI95 high** | 19.26820273037335 |
| **P95 latency** | 46.66529512485229 |
| **P99 latency** | 62.68256129697576 |
| **Deadline violation rate** | 0 |
| **Completion rate** | 1 |

## Network propagation delay ×2 | Local-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6472 |
| **Mean latency** | 20.129732177350427 |
| **Mean CI95 low** | 19.68368638697836 |
| **Mean CI95 high** | 20.57577796772249 |
| **P95 latency** | 95.89897420768688 |
| **P99 latency** | 120.4446636336917 |
| **Deadline violation rate** | 0.0585583006371256 |
| **Completion rate** | 1 |

## Network propagation delay ×2 | Cloud-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 6472 |
| **Mean latency** | 171.83233143324998 |
| **Mean CI95 low** | 171.2825582538571 |
| **Mean CI95 high** | 172.38210461264285 |
| **P95 latency** | 255.4815792152521 |
| **P99 latency** | 258.678867926011 |
| **Deadline violation rate** | 1 |
| **Completion rate** | 1 |

# v26 Results

Current v26 results — 2 scenarios, 5 seeds, 6 policy labels

## Heterogeneous task deadlines | Tabular Q-value / contextual-bandit policy

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1427 |
| **Mean latency** | 10.894380668454192 |
| **Mean CI95 low** | 10.178530462942833 |
| **Mean CI95 high** | 11.61023087396555 |
| **Mean latency SD** | 0.57652489445368 |
| **P95 latency** | 26.012549884144676 |
| **P99 latency** | 74.81930684325555 |
| **Deadline violation rate** | 0 |
| **Deadline CI95 low** | 0 |
| **Deadline CI95 high** | 0 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0 |

## Heterogeneous task deadlines | Latency-greedy heuristic

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1427 |
| **Mean latency** | 8.157663371960556 |
| **Mean CI95 low** | 7.661246902729409 |
| **Mean CI95 high** | 8.654079841191702 |
| **Mean latency SD** | 0.3997993579175051 |
| **P95 latency** | 24.333333333333332 |
| **P99 latency** | 41.00670784257155 |
| **Deadline violation rate** | 0 |
| **Deadline CI95 low** | 0 |
| **Deadline CI95 high** | 0 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0 |

## Heterogeneous task deadlines | Deadline-greedy diagnostic (duplicate)

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1427 |
| **Mean latency** | 8.157663371960556 |
| **Mean CI95 low** | 7.661246902729409 |
| **Mean CI95 high** | 8.654079841191702 |
| **Mean latency SD** | 0.3997993579175051 |
| **P95 latency** | 24.333333333333332 |
| **P99 latency** | 41.00670784257155 |
| **Deadline violation rate** | 0 |
| **Deadline CI95 low** | 0 |
| **Deadline CI95 high** | 0 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0 |

## Heterogeneous task deadlines | Edge-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1427 |
| **Mean latency** | 14.412070778405411 |
| **Mean CI95 low** | 13.642992545031431 |
| **Mean CI95 high** | 15.181149011779391 |
| **Mean latency SD** | 0.6193931969410051 |
| **P95 latency** | 40.582553150236755 |
| **P99 latency** | 62.23637625013955 |
| **Deadline violation rate** | 0.0013698630136986301 |
| **Deadline CI95 low** | -0.002433486445476429 |
| **Deadline CI95 high** | 0.00517321247287369 |
| **Completion rate** | 1 |
| **Small miss rate** | 0.001834862385321101 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0 |

## Heterogeneous task deadlines | Local-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1427 |
| **Mean latency** | 18.985495700859694 |
| **Mean CI95 low** | 18.351177232881774 |
| **Mean CI95 high** | 19.619814168837614 |
| **Mean latency SD** | 0.5108616090146351 |
| **P95 latency** | 75.22769421662164 |
| **P99 latency** | 119.21457481725808 |
| **Deadline violation rate** | 0.014027304986533456 |
| **Deadline CI95 low** | 0.002963004493876143 |
| **Deadline CI95 high** | 0.02509160547919077 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0.011320754716981133 |
| **Large miss rate** | 0.2111111111111111 |

## Heterogeneous task deadlines | Cloud-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1427 |
| **Mean latency** | 87.16555059731931 |
| **Mean CI95 low** | 86.46905680617947 |
| **Mean CI95 high** | 87.86204438845915 |
| **Mean latency SD** | 0.5609358023969603 |
| **P95 latency** | 130.07688489333046 |
| **P99 latency** | 133.33416367078664 |
| **Deadline violation rate** | 0.8809167276237201 |
| **Deadline CI95 low** | 0.8766352893234984 |
| **Deadline CI95 high** | 0.8851981659239417 |
| **Completion rate** | 1 |
| **Small miss rate** | 1 |
| **Medium miss rate** | 0.5166623376623376 |
| **Large miss rate** | 0.5 |

## Edge-1 compute-service outage/recovery | Tabular Q-value / contextual-bandit policy

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1441 |
| **Mean latency** | 12.982768513772692 |
| **Mean CI95 low** | 11.906966311935234 |
| **Mean CI95 high** | 14.05857071561015 |
| **Mean latency SD** | 0.866419742695087 |
| **P95 latency** | 34.19775836454335 |
| **P99 latency** | 106.19779552582392 |
| **Deadline violation rate** | 0.0020887228130688894 |
| **Deadline CI95 low** | -0.0018056694344086872 |
| **Deadline CI95 high** | 0.005983115060546466 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0.0375 |

## Edge-1 compute-service outage/recovery | Latency-greedy heuristic

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1441 |
| **Mean latency** | 11.272241557269842 |
| **Mean CI95 low** | 10.254683667657595 |
| **Mean CI95 high** | 12.289799446882089 |
| **Mean latency SD** | 0.8195114709649977 |
| **P95 latency** | 33.965975304931995 |
| **P99 latency** | 89.35615176933726 |
| **Deadline violation rate** | 0.0014084507042253522 |
| **Deadline CI95 low** | -0.002502035359433512 |
| **Deadline CI95 high** | 0.005318936767884216 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0.025 |

## Edge-1 compute-service outage/recovery | Deadline-greedy diagnostic (duplicate)

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1441 |
| **Mean latency** | 11.272241557269842 |
| **Mean CI95 low** | 10.254683667657595 |
| **Mean CI95 high** | 12.289799446882089 |
| **Mean latency SD** | 0.8195114709649977 |
| **P95 latency** | 33.965975304931995 |
| **P99 latency** | 89.35615176933726 |
| **Deadline violation rate** | 0.0014084507042253522 |
| **Deadline CI95 low** | -0.002502035359433512 |
| **Deadline CI95 high** | 0.005318936767884216 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0.025 |

## Edge-1 compute-service outage/recovery | Edge-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1441 |
| **Mean latency** | 15.249628123179685 |
| **Mean CI95 low** | 14.526347944531006 |
| **Mean CI95 high** | 15.972908301828364 |
| **Mean latency SD** | 0.5825087782967072 |
| **P95 latency** | 40.658505419640974 |
| **P99 latency** | 84.64975574349616 |
| **Deadline violation rate** | 0.0007042253521126761 |
| **Deadline CI95 low** | -0.001251017679716756 |
| **Deadline CI95 high** | 0.002659468383942108 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0 |
| **Large miss rate** | 0.0125 |

### Edge-1 compute-service outage/recovery | Local-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1441 |
| **Mean latency** | 18.84599103399635 |
| **Mean CI95 low** | 17.78377853735357 |
| **Mean CI95 high** | 19.908203530639128 |
| **Mean latency SD** | 0.8554749901578964 |
| **P95 latency** | 76.3455096146213 |
| **P99 latency** | 116.70864518316651 |
| **Deadline violation rate** | 0.0076505405750410815 |
| **Deadline CI95 low** | 0.004015874902322003 |
| **Deadline CI95 high** | 0.01128520624776016 |
| **Completion rate** | 1 |
| **Small miss rate** | 0 |
| **Medium miss rate** | 0.0035714285714285713 |
| **Large miss rate** | 0.12710084033613445 |

### Edge-1 compute-service outage/recovery | Cloud-only

| | |
|---|---|
| **Seeds** | 5 |
| **Tasks across 5 seeds** | 1441 |
| **Mean latency** | 87.02924228553805 |
| **Mean CI95 low** | 85.8247987908017 |
| **Mean CI95 high** | 88.2336857802744 |
| **Mean latency SD** | 0.9700236911747018 |
| **P95 latency** | 129.96860140423644 |
| **P99 latency** | 133.11310887563906 |
| **Deadline violation rate** | 0.8826269470856266 |
| **Deadline CI95 low** | 0.8700439891811476 |
| **Deadline CI95 high** | 0.8952099049901057 |
| **Completion rate** | 1 |
| **Small miss rate** | 1 |
| **Medium miss rate** | 0.5285004785613305 |
| **Large miss rate** | 0.5058823529411764 |

# Paired Differences

v26 paired differences — exploratory, n=5 seeds

### Heterogeneous task deadlines | qlearning minus latency_greedy | mean_latency

| | |
|---|---|
| **Mean paired difference** | 2.736717296493638 |
| **95% CI low** | 2.021303851333964 |
| **95% CI high** | 3.4521307416533125 |
| **SD paired difference** | 0.5761731403943557 |
| **Paired t p-value (exploratory)** | 0.0004449007400750203 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus latency_greedy | p95_latency

| | |
|---|---|
| **Mean paired difference** | 1.6792165508113441 |
| **95% CI low** | -1.2370960230837837 |
| **95% CI high** | 4.595529124706472 |
| **SD paired difference** | 2.3487131608180403 |
| **Paired t p-value (exploratory)** | 0.18513665377096733 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus latency_greedy | p99_latency

| | |
|---|---|
| **Mean paired difference** | 33.81259900068401 |
| **95% CI low** | 8.721046374486775 |
| **95% CI high** | 58.904151626881244 |
| **SD paired difference** | 20.208005275578238 |
| **Paired t p-value (exploratory)** | 0.020095703608662214 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus latency_greedy | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | 0 |
| **95% CI low** | 0 |
| **95% CI high** | 0 |
| **SD paired difference** | 0 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus deadline_greedy | mean_latency

| | |
|---|---|
| **Mean paired difference** | 2.736717296493638 |
| **95% CI low** | 2.021303851333964 |
| **95% CI high** | 3.4521307416533125 |
| **SD paired difference** | 0.5761731403943557 |
| **Paired t p-value (exploratory)** | 0.0004449007400750203 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus deadline_greedy | p95_latency

| | |
|---|---|
| **Mean paired difference** | 1.6792165508113441 |
| **95% CI low** | -1.2370960230837837 |
| **95% CI high** | 4.595529124706472 |
| **SD paired difference** | 2.3487131608180403 |
| **Paired t p-value (exploratory)** | 0.18513665377096733 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus deadline_greedy | p99_latency

| | |
|---|---|
| **Mean paired difference** | 33.81259900068401 |
| **95% CI low** | 8.721046374486775 |
| **95% CI high** | 58.904151626881244 |
| **SD paired difference** | 20.208005275578238 |
| **Paired t p-value (exploratory)** | 0.020095703608662214 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus deadline_greedy | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | 0 |
| **95% CI low** | 0 |
| **95% CI high** | 0 |
| **SD paired difference** | 0 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus edge | mean_latency

| | |
|---|---|
| **Mean paired difference** | -3.517690109951219 |
| **95% CI low** | -4.2590240640050645 |
| **95% CI high** | -2.7763561558973735 |
| **SD paired difference** | 0.5970487628909961 |
| **Paired t p-value (exploratory)** | 0.00019174339458380974 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus edge | p95_latency

| | |
|---|---|
| **Mean paired difference** | -14.570003266092076 |
| **95% CI low** | -19.13173556643243 |
| **95% CI high** | -10.008270965751723 |
| **SD paired difference** | 3.6738862582305023 |
| **Paired t p-value (exploratory)** | 0.0008931542709046567 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus edge | p99_latency

| | |
|---|---|
| **Mean paired difference** | 12.582930593116009 |
| **95% CI low** | -9.7997711099932 |
| **95% CI high** | 34.965632296225216 |
| **SD paired difference** | 18.02637568254281 |
| **Paired t p-value (exploratory)** | 0.19358260552259987 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus edge | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | -0.0013698630136986301 |
| **95% CI low** | -0.00517321247287369 |
| **95% CI high** | 0.002433486445476429 |
| **SD paired difference** | 0.0030631068184928624 |
| **Paired t p-value (exploratory)** | 0.37390096630005887 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus local | mean_latency

| | |
|---|---|
| **Mean paired difference** | -8.091115032405506 |
| **95% CI low** | -8.811656406088794 |
| **95% CI high** | -7.370573658722218 |
| **SD paired difference** | 0.5803030245909112 |
| **Paired t p-value (exploratory)** | 6.307021441510981e-06 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus local | p95_latency

| | |
|---|---|
| **Mean paired difference** | -49.215144332476974 |
| **95% CI low** | -60.48440370420216 |
| **95% CI high** | -37.94588496075179 |
| **SD paired difference** | 9.07593309303298 |
| **Paired t p-value (exploratory)** | 0.0002654243430932459 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus local | p99_latency

| | |
|---|---|
| **Mean paired difference** | -44.39526797400252 |
| **95% CI low** | -62.76151453601217 |
| **95% CI high** | -26.02902141199287 |
| **SD paired difference** | 14.791639758081837 |
| **Paired t p-value (exploratory)** | 0.002565872738415208 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus local | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | -0.014027304986533456 |
| **95% CI low** | -0.02509160547919077 |
| **95% CI high** | -0.002963004493876143 |
| **SD paired difference** | 0.008910865184674219 |
| **Paired t p-value (exploratory)** | 0.024450640957690364 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus cloud | mean_latency

| | |
|---|---|
| **Mean paired difference** | -76.27116992886513 |
| **95% CI low** | -77.25114668633213 |
| **95% CI high** | -75.29119317139813 |
| **SD paired difference** | 0.7892447223118921 |
| **Paired t p-value (exploratory)** | 2.751403926548937e-09 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus cloud | p95_latency

| | |
|---|---|
| **Mean paired difference** | -104.06433500918577 |
| **95% CI low** | -107.56655326810707 |
| **95% CI high** | -100.56211675026447 |
| **SD paired difference** | 2.8205845252723303 |
| **Paired t p-value (exploratory)** | 1.2940015115470773e-07 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus cloud | p99_latency

| | |
|---|---|
| **Mean paired difference** | -58.51485682753109 |
| **95% CI low** | -83.96299051411276 |
| **95% CI high** | -33.06672314094943 |
| **SD paired difference** | 20.495185270246903 |
| **Paired t p-value (exploratory)** | 0.003089272414453366 |
| **Paired seeds** | 5 |

## Heterogeneous task deadlines | qlearning minus cloud | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | -0.8809167276237201 |
| **95% CI low** | -0.8851981659239417 |
| **95% CI high** | -0.8766352893234984 |
| **SD paired difference** | 0.0034481456387681227 |
| **Paired t p-value (exploratory)** | 5.633850851386286e-11 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus latency_greedy | mean_latency

| | |
|---|---|
| **Mean paired difference** | 1.7105269565028496 |
| **95% CI low** | 1.3279110691701912 |
| **95% CI high** | 2.093142843835508 |
| **SD paired difference** | 0.3081476855946249 |
| **Paired t p-value (exploratory)** | 0.000242194755116095 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus latency_greedy | p95_latency

| | |
|---|---|
| **Mean paired difference** | 0.2317830596113609 |
| **95% CI low** | -1.2685891339088873 |
| **95% CI high** | 1.732155253131609 |
| **SD paired difference** | 1.2083560413209535 |
| **Paired t p-value (exploratory)** | 0.6900751267903941 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus latency_greedy | p99_latency

| | |
|---|---|
| **Mean paired difference** | 16.841643756486658 |
| **95% CI low** | -8.96251676523963 |
| **95% CI high** | 42.64580427821295 |
| **SD paired difference** | 20.781918908058476 |
| **Paired t p-value (exploratory)** | 0.14419365163485887 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus latency_greedy | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | 0.0006802721088435374 |
| **95% CI low** | -0.0012084660579576825 |
| **95% CI high** | 0.002569010275644757 |
| **SD paired difference** | 0.0015211346785712853 |
| **Paired t p-value (exploratory)** | 0.37390096630005887 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus deadline_greedy | mean_latency

| | |
|---|---|
| **Mean paired difference** | 1.7105269565028496 |
| **95% CI low** | 1.3279110691701912 |
| **95% CI high** | 2.093142843835508 |
| **SD paired difference** | 0.3081476855946249 |
| **Paired t p-value (exploratory)** | 0.000242194755116095 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus deadline_greedy | p95_latency

| | |
|---|---|
| **Mean paired difference** | 0.2317830596113609 |
| **95% CI low** | -1.2685891339088873 |
| **95% CI high** | 1.732155253131609 |
| **SD paired difference** | 1.2083560413209535 |
| **Paired t p-value (exploratory)** | 0.6900751267903941 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus deadline_greedy | p99_latency

| | |
|---|---|
| **Mean paired difference** | 16.841643756486658 |
| **95% CI low** | -8.96251676523963 |
| **95% CI high** | 42.64580427821295 |
| **SD paired difference** | 20.781918908058476 |
| **Paired t p-value (exploratory)** | 0.14419365163485887 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus deadline_greedy | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | 0.0006802721088435374 |
| **95% CI low** | -0.0012084660579576825 |
| **95% CI high** | 0.002569010275644757 |
| **SD paired difference** | 0.0015211346785712853 |
| **Paired t p-value (exploratory)** | 0.37390096630005887 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus edge | mean_latency

| | |
|---|---|
| **Mean paired difference** | -2.266859609406991 |
| **95% CI low** | -2.8379873847348227 |
| **95% CI high** | -1.6957318340791587 |
| **SD paired difference** | 0.4599696666360997 |
| **Paired t p-value (exploratory)** | 0.0003854397157801016 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus edge | p95_latency

| | |
|---|---|
| **Mean paired difference** | -6.4607470550976185 |
| **95% CI low** | -8.666572523821106 |
| **95% CI high** | -4.254921586374131 |
| **SD paired difference** | 1.7765075510883086 |
| **Paired t p-value (exploratory)** | 0.0012439066431837503 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus edge | p99_latency

| | |
|---|---|
| **Mean paired difference** | 21.54803978232776 |
| **95% CI low** | 13.439393835153378 |
| **95% CI high** | 29.65668572950214 |
| **SD paired difference** | 6.530467146429824 |
| **Paired t p-value (exploratory)** | 0.0017986992806820249 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus edge | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | 0.0013844974609562135 |
| **95% CI low** | -0.0009700377599430178 |
| **95% CI high** | 0.0037390326818554447 |
| **SD paired difference** | 0.0018962740518412282 |
| **Paired t p-value (exploratory)** | 0.17789298738170595 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus local | mean_latency

| | |
|---|---|
| **Mean paired difference** | -5.8632225202236565 |
| **95% CI low** | -6.331517892217525 |
| **95% CI high** | -5.3949271482297885 |
| **SD paired difference** | 0.3771514456981284 |
| **Paired t p-value (exploratory)** | 4.0863603462369705e-06 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus local | p95_latency

| | |
|---|---|
| **Mean paired difference** | -42.147751250077945 |
| **95% CI low** | -65.8161747160434 |
| **95% CI high** | -18.47932778411249 |
| **SD paired difference** | 19.061858522277404 |
| **Paired t p-value (exploratory)** | 0.007793498586263375 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus local | p99_latency

| | |
|---|---|
| **Mean paired difference** | -10.510849657342593 |
| **95% CI low** | -21.23487957068243 |
| **95% CI high** | 0.21318025599724244 |
| **SD paired difference** | 8.636821176142341 |
| **Paired t p-value (exploratory)** | 0.05291836582520029 |
| **Paired seeds** | 5 |

## Edge-1 compute-service outage/recovery | qlearning minus local | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | -0.005561817761972192 |
| **95% CI low** | -0.010430665078627277 |
| **95% CI high** | -0.0006929704453171063 |
| **SD paired difference** | 0.003921227742528201 |
| **Paired t p-value (exploratory)** | 0.033805638285807856 |
| **Paired seeds** | 5 |

### Edge-1 compute-service outage/recovery | qlearning minus cloud | mean_latency

| | |
|---|---|
| **Mean paired difference** | -74.04647377176535 |
| **95% CI low** | -75.70169498665756 |
| **95% CI high** | -72.39125255687314 |
| **SD paired difference** | 1.3330669305759908 |
| **Paired t p-value (exploratory)** | 2.5200773971793594e-08 |
| **Paired seeds** | 5 |

### Edge-1 compute-service outage/recovery | qlearning minus cloud | p95_latency

| | |
|---|---|
| **Mean paired difference** | -95.77084303969309 |
| **95% CI low** | -98.45288905135995 |
| **95% CI high** | -93.08879702802624 |
| **SD paired difference** | 2.160041698516505 |
| **Paired t p-value (exploratory)** | 6.206289546138609e-08 |
| **Paired seeds** | 5 |

### Edge-1 compute-service outage/recovery | qlearning minus cloud | p99_latency

| | |
|---|---|
| **Mean paired difference** | -26.915313349815165 |
| **95% CI low** | -38.568402245572734 |
| **95% CI high** | -15.262224454057597 |
| **SD paired difference** | 9.385058206258172 |
| **Paired t p-value (exploratory)** | 0.003038391888610079 |
| **Paired seeds** | 5 |

### Edge-1 compute-service outage/recovery | qlearning minus cloud | deadline_violation_rate

| | |
|---|---|
| **Mean paired difference** | -0.8805382242725578 |
| **95% CI low** | -0.8954113199669449 |
| **95% CI high** | -0.8656651285781707 |
| **SD paired difference** | 0.01197835784552269 |
| **Paired t p-value (exploratory)** | 8.216755629132049e-09 |
| **Paired seeds** | 5 |

# Legacy Seed1-2

Earlier technical audit — archival context only

### 1 | Frozen RL

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 224 |
| **Pending** | 4 |
| **Completion rate** | 0.9825 |
| **Average True E2E** | 11.1952 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 1 | Local-only

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 218 |
| **Pending** | 10 |
| **Completion rate** | 0.9561 |
| **Average True E2E** | 17.4009 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 1 | Edge-only

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 222 |
| **Pending** | 6 |
| **Completion rate** | 0.9737 |
| **Average True E2E** | 14.2531 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 1 | Cloud-only

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 183 |
| **Pending** | 45 |
| **Completion rate** | 0.8026 |
| **Average True E2E** | 81.6142 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 2 | Frozen RL

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 222 |
| **Pending** | 6 |
| **Completion rate** | 0.9737 |
| **Average True E2E** | 9.9644 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 2 | Local-only

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 215 |
| **Pending** | 13 |
| **Completion rate** | 0.943 |
| **Average True E2E** | 17.2413 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 2 | Edge-only

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 218 |
| **Pending** | 10 |
| **Completion rate** | 0.9561 |
| **Average True E2E** | 13.711 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

## 2 | Cloud-only

| | |
|---|---|
| **Audit snapshot** | Member 3 earlier audit |
| **Generated** | 228 |
| **Completed** | 178 |
| **Pending** | 50 |
| **Completion rate** | 0.7807 |
| **Average True E2E** | 82.2194 |
| **Measurement cutoff** | 500 |
| **Deadline setting** | Fixed global deadline 500 |
| **Status** | Legacy only; do not merge with v26 |

# Research Gap v29

Targeted research-gap matrix v29 — 21 records; not systematic

## Forecast-driven task offloading for reliable and adaptive mobile edge computing

| | |
|---|---|
| **ID** | SD-01 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.jnca.2026.104494 |
| **Link** | https://doi.org/10.1016/j.jnca.2026.104494 |
| **Methodology** | TimesNet multi-step forecast of server load and failure dynamics + LinUCB contextual-bandit offloading; real-world multivariate traces and simulated failure processes. |
| **Deadline / risk treatment** | Reliability-aware and failure-aware using forecasted context; abstract does not establish a formal per-task chance constraint. |
| **Baselines / metrics evidenced** | Publisher record reports lower task failures/system cost and faster convergence; full exact baseline table not extracted in this audit. |
| **Overlap with our project** | Very high for contextual bandit + adaptive offloading + reliability/failure context; forecasting is not implemented in our project. |
| **Defensible distinction / implication** | Only a narrow empirical distinction could be claimed if our contribution is specifically a reproducible iFogSim2 comparison of a simple tabular contextual policy against matched heuristics. Do not claim generic forecast-aware or reliability-aware bandit novelty. |
| **Evidence grade** | Publisher full page/abstract and indexed algorithm text reviewed; exact full equations/tables not independently transcribed. |
| **Publisher evidence** | Official ScienceDirect page |

## RELTO: A reliability-oriented DRL approach with context-aware adaptive reward weighting for multi-objective task offloading in MEC

| | |
|---|---|
| **ID** | SD-02 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.adhoc.2025.104065 |
| **Link** | https://doi.org/10.1016/j.adhoc.2025.104065 |
| **Methodology** | PPO with hybrid action space and dual-attention mechanism for context-aware adaptive weighting of reliability, latency, and energy objectives. |
| **Deadline / risk treatment** | Reliability explicitly optimized alongside latency and energy; exact formal guarantee not verified from accessible record. |
| **Baselines / metrics evidenced** | Publisher abstract/highlights report reliability, energy and latency comparisons; complete baseline table not extracted. |
| **Overlap with our project** | High for reliability-aware offloading and context-sensitive objective weighting; current project has a fixed weighted reward and no adaptive weights. |
| **Defensible distinction / implication** | A fixed tabular reward is a simpler baseline, not a new adaptive risk mechanism. Any new claim must identify a different mathematically specified mechanism and compare fairly. |
| **Evidence grade** | Publisher abstract/highlights and indexed text; full equations/tables not independently extracted. |
| **Publisher evidence** | Official ScienceDirect page |

## Chance constraint-based dependent task offloading for multi-access edge computing under uncertainty

| | |
|---|---|
| **ID** | SD-03 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.asoc.2026.115710 |
| **Link** | https://doi.org/10.1016/j.asoc.2026.115710 |
| **Methodology** | Uncertain joint optimization for dependent tasks; decomposed chance-constrained subproblems solved with improved genetic algorithm, Monte Carlo simulation, and ANN approximation. |
| **Deadline / risk treatment** | Explicit chance constraints under stochastic uncertainty; jointly addresses latency, energy, task utility, caching, multipath transmission, and privacy. |
| **Baselines / metrics evidenced** | Publisher abstract reports comparison with five algorithms; exact names/table not extracted. |
| **Overlap with our project** | High for the broad claim of uncertainty-aware/chance-constrained task offloading, but the modeled problem differs substantially from independent local/edge/cloud actions. |
| **Defensible distinction / implication** | Our current binary deadline-miss penalty is not a chance constraint. Do not claim probabilistic reliability unless a probability model, calibration, and constraint satisfaction evaluation are implemented. |
| **Evidence grade** | Official publisher abstract and section snippets; full equations/table not independently extracted. |
| **Publisher evidence** | Official ScienceDirect page |

## Deadline-Aware joint optimization of task offloading and resource allocation for cell-Free mobile edge computing networks

| | |
|---|---|
| **ID** | SD-04 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.comnet.2026.112219 |
| **Link** | https://doi.org/10.1016/j.comnet.2026.112219 |
| **Methodology** | JAB-TORA integrates BS–UE association, beam design, task split ratio, and resource allocation; MADRL is used for dynamic channel/power decisions. |
| **Deadline / risk treatment** | Objective minimizes a weighted sum of average delay and deadline-related dropout ratio. |
| **Baselines / metrics evidenced** | Publisher record describes comparisons with representative optimization- and learning-based benchmarks; exact names not fully extracted. |
| **Overlap with our project** | High for deadline-aware offloading and deadline/dropout metrics; distinct cell-free wireless architecture and joint radio-resource decisions. |
| **Defensible distinction / implication** | Deadline-aware objective and violation/dropout reporting are not novel alone. Our distinct scope must be stated as an iFogSim2 edge–cloud empirical protocol, not a new deadline-aware principle. |
| **Evidence grade** | Official ScienceDirect abstract/highlights and indexed sections. |
| **Publisher evidence** | Official ScienceDirect page |

## Joint offloading and energy optimization in NOMA-aided UAV-MEC under practical impairments: A continuous contextual bandit DRL framework

| | |
|---|---|
| **ID** | SD-05 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.phycom.2026.103309 |
| **Link** | https://doi.org/10.1016/j.phycom.2026.103309 |
| **Methodology** | Continuous contextual bandit Soft Actor-Critic (CCB-SAC), gamma=0, jointly optimizes UAV 3D position, power-allocation ratio, and task-offloading ratio using analytical probability boundaries as reward. |
| **Deadline / risk treatment** | Derives successful-computation probability and energy-outage probability under imperfect CSI/SIC; quasi-static environment is explicitly reformulated as a contextual bandit. |
| **Baselines / metrics evidenced** | CHEA, EHGA, ABC, PSO and standard MDP-based DRL; successful computation probability, energy outage, optimality gap and inference complexity. |
| **Overlap with our project** | Very high methodological warning: contextual single-step learning with gamma=0 is explicitly used in a 2026 offloading paper. System setting differs (UAV/NOMA/continuous joint control). |
| **Defensible distinction / implication** | The fact that gamma=0 can be appropriate in a quasi-static contextual problem is not novel. We must justify whether our sequential simulator really has no delayed consequences; otherwise use gamma>0 and test temporal credit assignment. |
| **Evidence grade** | Official ScienceDirect full-length page/indexed full text reviewed; DOI and abstract verified. |
| **Publisher evidence** | Official ScienceDirect page |

## A fault-tolerant task offloading framework via large-scale multi-objective evolutionary optimization and game-based decision mechanism

| | |
|---|---|
| **ID** | SD-06 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.eswa.2025.129910 |
| **Link** | https://doi.org/10.1016/j.eswa.2025.129910 |
| **Methodology** | ALC-LSMOEA-KT large-scale multi-objective evolutionary optimization with knowledge transfer, plus Stackelberg game-based migration for recovery. |
| **Deadline / risk treatment** | Jointly models delay, energy, load balance, and risk; fault-tolerant recovery in dynamic MEC. |
| **Baselines / metrics evidenced** | Publisher abstract describes multi-objective optimization and fault-tolerant comparisons; complete baseline table not extracted. |
| **Overlap with our project** | Adjacent prior art for failure-aware/fault-tolerant offloading; algorithm family differs. |
| **Defensible distinction / implication** | A generic claim of handling failures is not novel. If testing failure scenarios, define failure semantics and report them as comparative robustness experiments. |
| **Evidence grade** | Official ScienceDirect abstract/highlights. |
| **Publisher evidence** | Official ScienceDirect page |

## DELTA: Deadline aware energy and latency-optimized task offloading and resource allocation in GPU-enabled, PiM-enabled distributed heterogeneous MEC architecture

| | |
|---|---|
| **ID** | SD-07 |
| **Database** | ScienceDirect |
| **Year** | 2025 |
| **DOI** | 10.1016/j.sysarc.2025.103335 |
| **Link** | https://doi.org/10.1016/j.sysarc.2025.103335 |
| **Methodology** | Multi-objective mixed-integer programming formulation and deadline-aware energy/latency offloading/resource allocation in a heterogeneous cooperative MEC architecture. |
| **Deadline / risk treatment** | Explicitly handles application deadlines and task dependencies while minimizing UE energy and application latency. |
| **Baselines / metrics evidenced** | Publisher abstract lists Intelligent-TO, Multi-user, and Selective-random policies; latency and UE energy. |
| **Overlap with our project** | High for deadline-aware latency optimization, although it is optimization-based and includes GPU/SSD/PiM resources and dependent applications. |
| **Defensible distinction / implication** | Deadline-aware task offloading is established. A defensible distinction must be in the exact experimental protocol or a genuinely different algorithm, not deadline inclusion alone. |
| **Evidence grade** | Official ScienceDirect abstract. |
| **Publisher evidence** | Official ScienceDirect page |

## Multi-armed bandit-based federated reinforcement learning for dynamic task offloading in a containerized MEC

| | |
|---|---|
| **ID** | SPR-01 |
| **Database** | SpringerLink |
| **Year** | 2026 |
| **DOI** | 10.1007/s42452-026-08386-7 |
| **Link** | https://doi.org/10.1007/s42452-026-08386-7 |
| **Methodology** | Hybrid MAB for lightweight device-layer real-time decisions plus federated reinforcement learning across edge servers in a containerized MEC environment. |
| **Deadline / risk treatment** | Focuses on dynamic workloads, QoE, latency and energy; no explicit formal chance constraint identified in the reviewed abstract. |
| **Baselines / metrics evidenced** | Publisher reports comparisons with local/greedy offloading and centralized DRL; QoE, latency, energy. |
| **Overlap with our project** | High for MAB-based dynamic offloading; adds federated coordination and containerized deployment absent from our project. |
| **Defensible distinction / implication** | A bandit for dynamic task offloading is already prior art. Our only plausible difference is a tightly scoped, reproducible iFogSim2 comparison under matched baselines and task semantics. |
| **Evidence grade** | Official SpringerLink open-access full text/abstract reviewed. |
| **Publisher evidence** | Official SpringerLink page |

## Dynamic task offloading strategy in mobile edge computing using meta-reinforcement learning with adaptive learning rate adjustment

| | |
|---|---|
| **ID** | SPR-02 |
| **Database** | SpringerLink |
| **Year** | 2026 |
| **DOI** | 10.1007/s44443-026-00736-0 |
| **Link** | https://doi.org/10.1007/s44443-026-00736-0 |
| **Methodology** | ALR-MRL: meta-reinforcement learning with Seq2Seq task-graph embeddings and adaptive learning rates based on gradient variance/bias in a cloud–edge–end architecture. |
| **Deadline / risk treatment** | Optimizes latency on synthetic DAG workflows and varying resources/bandwidth; no formal chance constraint established in the inspected article sections. |
| **Baselines / metrics evidenced** | Greedy, fine-tuning DRL, MRLCO and HEFT-like priority scheduling; latency across resource/bandwidth groups. |
| **Overlap with our project** | Adjacent for adaptive offloading under changing resources, three-tier execution, and task-graph state; more complex temporal/meta-learning method. |
| **Defensible distinction / implication** | Resource-shift testing is useful but not novel by itself. If our task model has no DAG dependencies, that is a scope difference, not an algorithmic innovation. |
| **Evidence grade** | Official SpringerLink full text reviewed. |
| **Publisher evidence** | Official SpringerLink page |

## Computation offloading based on incomplete information in edge computing networks

| | |
|---|---|
| **ID** | SPR-03 |
| **Database** | SpringerLink |
| **Year** | 2025 |
| **DOI** | 10.1007/s10586-025-05560-1 |
| **Link** | https://doi.org/10.1007/s10586-025-05560-1 |
| **Methodology** | Distributed auction multi-armed bandit (DA-MAB/TODAuction-MAB) plus heterogeneous distributed MAB (HD-MAB); sliding-window adaptation to non-stationary rewards and regret analysis. |
| **Deadline / risk treatment** | Latency-sensitive allocation subject to network latency and incomplete/local feedback; no explicit per-task probabilistic deadline guarantee in the abstracted model. |
| **Baselines / metrics evidenced** | Simulation studies of latency-sensitive rewards, average delay, resource utilization, server heterogeneity and regret. |
| **Overlap with our project** | High for online bandit offloading under incomplete state and non-stationary network/server load. |
| **Defensible distinction / implication** | Bandit adaptation to dynamic latency is already established. A claim must not be based only on discretized context or a changing workload. |
| **Evidence grade** | Official SpringerLink open-access full text reviewed; system model and regret discussion inspected. |
| **Publisher evidence** | Official SpringerLink page |

## Time-and-Traffic-aware collaborative task offloading with service caching-replacement in cloud-assisted mobile edge computing

| | |
|---|---|
| **ID** | SPR-04 |
| **Database** | SpringerLink |
| **Year** | 2025 |
| **DOI** | 10.1007/s10586-025-05629-x |
| **Link** | https://doi.org/10.1007/s10586-025-05629-x |
| **Methodology** | Irregular Cellular Learning Automata classify traffic patterns/timescales; distributed deep RL for decentralized offloading; dynamic 0–1 knapsack for service caching; Q-learning for service replacement. |
| **Deadline / risk treatment** | Time/traffic-aware latency and energy optimization; not a formal chance-constrained reliability method. |
| **Baselines / metrics evidenced** | Publisher abstract reports comparison against traditional centralized approaches; detailed baseline table should be extracted if this paper is retained as a core comparator. |
| **Overlap with our project** | High for Q-learning and cloud-assisted edge offloading under changing traffic; caching/replacement is outside our current scope. |
| **Defensible distinction / implication** | Q-learning and traffic-aware cloud-edge offloading are established. Our possible distinction is a smaller controlled benchmark, not a new Q-learning/offloading concept. |
| **Evidence grade** | Official SpringerLink open-access full text/abstract reviewed. |
| **Publisher evidence** | Official SpringerLink page |

## Task Offloading and Resource Optimization Based on Dependency-Aware Graph and Collaborative Deep Reinforcement Learning in Mobile Edge Computing

| | |
|---|---|
| **ID** | IEEE-01 |
| **Database** | IEEE Xplore |
| **Year** | 2026 |
| **DOI** | 10.1109/TMC.2026.3665729 |
| **Link** | https://doi.org/10.1109/TMC.2026.3665729 |
| **Methodology** | Dependency-aware graph feature extraction plus graph-based multi-agent DRL (DGMA), adaptive prioritized replay, and correlation-based selective parameter sharing. |
| **Deadline / risk treatment** | Deadline violation rate is an explicit reported metric; handles dependent tasks, dynamic network conditions and partial observability. |
| **Baselines / metrics evidenced** | IEEE metadata/abstract report delay, energy, offloading utility and deadline violation rate; exact baseline table requires authorized full text. |
| **Overlap with our project** | High for RL offloading and deadline-violation evaluation; task dependency and multi-agent graph learning are not in our current model. |
| **Defensible distinction / implication** | Deadline-miss reporting and RL offloading are not novel. Our setting is simpler; comparison should acknowledge the different task-dependency/agent architecture. |
| **Evidence grade** | Official IEEE Xplore metadata/abstract; full paper not available in this audit. |
| **Publisher evidence** | Official IEEE Xplore page |

## CVaR-Constrained Reinforcement Learning for Ultra-Reliable Task Offloading in LEO Satellite Edge Computing

| | |
|---|---|
| **ID** | IEEE-02 |
| **Database** | IEEE Xplore |
| **Year** | 2026 |
| **DOI** | 10.1109/PRMVAI70103.2026.11605648 |
| **Link** | https://doi.org/10.1109/PRMVAI70103.2026.11605648 |
| **Methodology** | CVaR-constrained risk-aware reinforcement learning for task offloading in a dynamic LEO satellite edge-computing system. |
| **Deadline / risk treatment** | Explicitly constrains tail risk via CVaR to improve reliability beyond expected-latency optimization. |
| **Baselines / metrics evidenced** | IEEE abstract states latency and reliability comparisons against baselines; exact algorithms and numerical tables not extracted. |
| **Overlap with our project** | High for the broad claim that risk-aware/catastrophic-tail offloading is novel; application setting is satellite edge rather than iFogSim2 edge-cloud. |
| **Defensible distinction / implication** | Do not claim CVaR/tail-risk-aware offloading as a new general idea. Our current model does not implement CVaR at all. |
| **Evidence grade** | Official IEEE Xplore abstract/metadata; full paper not extracted. |
| **Publisher evidence** | Official IEEE Xplore page |

## Risk-Aware Graph-Collaborative Deep Reinforcement Learning for UAV-Enabled MEC Task Offloading

| | |
|---|---|
| **ID** | IEEE-03 |
| **Database** | IEEE Xplore (metadata found; secondary index used for DOI cross-check) |
| **Year** | 2026 |
| **DOI** | 10.1109/ICETCI68937.2026.11610771 |
| **Link** | https://doi.org/10.1109/ICETCI68937.2026.11610771 |
| **Methodology** | Graph-collaborative DRL with a distributional value critic and CVaR regularization; hybrid discrete/continuous offloading/resource actions. |
| **Deadline / risk treatment** | Explicit tail-risk suppression under stochastic traffic, channel blockage and energy constraints. |
| **Baselines / metrics evidenced** | Secondary bibliographic record describes the method; full official IEEE record and baseline tables not independently accessed. |
| **Overlap with our project** | High for generic risk-aware DRL/CVaR claims, but UAV-specific action space differs. |
| **Defensible distinction / implication** | Keep as a supplementary prior-art alert with evidence grade limited; do not rely on it for detailed equations until official full record/text is obtained. |
| **Evidence grade** | Secondary bibliographic record/abstract only; official full text not verified. |
| **Publisher evidence** | DOI metadata cross-checked using secondary index |

## Energy-Efficient Deadline-Aware Edge Computing (E2DA)

| | |
|---|---|
| **ID** | PRIOR-01 |
| **Database** | IEEE Xplore / IEEE conference record |
| **Year** | 2023 |
| **DOI** | 10.1109/GLOBECOM54140.2023.10437602 |
| **Link** | https://arxiv.org/html/2308.06647 |
| **Methodology** | Neural epsilon-greedy contextual multi-armed bandit; context includes task size, computational intensity, and deadline; action is local processing or offloading via a wireless channel. |
| **Deadline / risk treatment** | Deadline-aware reward uses an indicator for meeting the deadline; optimizes energy efficiency and response time. |
| **Baselines / metrics evidenced** | Oracle-style EEL*, EE*, R*; response time and energy efficiency; ns-3/LTE/MEC simulation. |
| **Overlap with our project** | Very high for contextual-bandit + deadline-aware reward, although the system/network and objective differ. |
| **Defensible distinction / implication** | A deadline feature and miss penalty in a contextual bandit cannot be claimed as novel alone. |
| **Evidence grade** | Full preprint text reviewed. |
| **Publisher evidence** | Open full-text preprint; DOI verified |

## Chance-Constrained Task Offloading for Reliability Guarantees in Multi-Tenant Networks (O2C/CCMAB)

| | |
|---|---|
| **ID** | PRIOR-02 |
| **Database** | IEEE ICC / user-provided paper record |
| **Year** | 2026 |
| **DOI** | 10.1109/ICC59461.2026.11586921 |
| **Link** | https://hal.science/hal-05705287v1 |
| **Methodology** | Chance-constrained multi-tenant server assignment; KL-based upper-confidence estimates of feasible co-tenancy/success probability; integer linear program for allocation. |
| **Deadline / risk treatment** | Explicit constraint P(delay <= task latency limit) >= target reliability; derives admissibility and regret guarantees. |
| **Baselines / metrics evidenced** | Reliability and regret under soft/strict settings; full PDF was previously inspected in project work. |
| **Overlap with our project** | Very high for any proposed claim about probabilistic deadline reliability or chance-constrained offloading. |
| **Defensible distinction / implication** | Current binary reward penalty is not a chance constraint and offers no probabilistic guarantee; a new formulation would need a distinct constraint/model and direct comparison. |
| **Evidence grade** | User-provided full PDF previously reviewed; HAL record/link available. |

## Lemur-Inspired Task Offloading for Edge–Fog–Cloud Continuum Systems (LITO)

| | |
|---|---|
| **ID** | PRIOR-03 |
| **Database** | MDPI / supplementary prior art |
| **Year** | 2026 |
| **DOI** | 10.3390/s26051497 |
| **Link** | https://www.mdpi.com/1424-8220/26/5/1497 |
| **Methodology** | Hierarchical VFDT local/offload decision, contextual CSPL-CB support-node selection, EDF–DRS scheduling, feasibility filter and fallback. |
| **Deadline / risk treatment** | Heterogeneous deadlines, queue/resource feasibility, safe fallback, node deactivation and redistribution experiments. |
| **Baselines / metrics evidenced** | Latency, SLA violations, task success, energy, utilization, throughput/fairness, ablations. |
| **Overlap with our project** | Very high for edge–fog–cloud, heterogeneous deadlines, contextual bandit, node failure and fallback. |
| **Defensible distinction / implication** | Deadline classes and outage tests alone are not novel; LITO should be in the mandatory equation/experiment comparison. |
| **Evidence grade** | Full text/indexed equations and algorithms reviewed in previous audit. |

## QoS-Aware Task Offloading in Fog-IoT Using Hybrid Deep Q-Learning with Actor-Critic (H-DQAC)

| | |
|---|---|
| **ID** | PRIOR-04 |
| **Database** | IIETA / supplementary prior art |
| **Year** | 2025 |
| **DOI** | 10.18280/jesa.581016 |
| **Link** | https://iieta.org/journals/jesa/paper/10.18280/jesa.581016 |
| **Methodology** | Hybrid DQL/Double-DQN and Actor-Critic; local/fog/cloud actions; iFogSim2 evaluation. |
| **Deadline / risk treatment** | Deadline-miss count is part of reward, alongside normalized latency and energy. |
| **Baselines / metrics evidenced** | Q-learning, DQN, multi-agent DRL, federated DRL; latency, task success, deadline misses, energy, utilization, throughput/QoE. |
| **Overlap with our project** | Very high for same simulator family, action set, deadline-aware reward and RL offloading. |
| **Defensible distinction / implication** | Using iFogSim2 or Q-learning with a deadline penalty is not novel by itself. |

## Fuzzy chance constraint-based task offloading for MEC under uncertainties with joint multi-network collaboration and privacy preservation

| | |
|---|---|
| **ID** | SPR-05 |
| **Database** | SpringerLink |
| **Year** | 2026 |
| **DOI** | 10.1007/s44443-026-00567-z |
| **Link** | https://link.springer.com/article/10.1007/s44443-026-00567-z |
| **Methodology** | Fuzzy optimization model transformed into fuzzy chance-constrained optimization with predefined confidence levels; hybrid improved adaptive genetic algorithm + Monte Carlo simulation + neural-network surrogate evaluation. |
| **Deadline / risk treatment** | Explicit fuzzy chance constraints for uncertainty/privacy; jointly optimizes latency, energy, multi-network data allocation and privacy. Not a contextual-bandit method. |
| **Baselines / metrics evidenced** | Springer full text states benchmarking against four algorithms; reports at least 23.24% average objective-value reduction. Exact baseline names should be extracted from the comparison table if used as a core baseline. |
| **Overlap with our project** | Relevant to uncertainty-aware and chance-constrained offloading, but its mathematical formulation and algorithm family differ from our tabular contextual policy. |
| **Defensible distinction / implication** | Current project does not implement fuzzy chance constraints or privacy constraints. This paper is evidence that uncertainty-aware offloading is prior art, not that the same problem is solved. |
| **Evidence grade** | Official SpringerLink full text reviewed; system model and methodology sections accessible. |
| **Publisher evidence** | Official SpringerLink full text |

### Lyapunov Optimization with Virtual Queues for adaptive task offloading in multi access edge computing

| | |
|---|---|
| **ID** | SPR-06 |
| **Database** | SpringerLink |
| **Year** | 2026 |
| **DOI** | 10.1007/s10791-026-10660-6 |
| **Link** | https://link.springer.com/article/10.1007/s10791-026-10660-6 |
| **Methodology** | Lyapunov Drift-Plus-Penalty with virtual energy queues, decay-aware queue updates and battery-aware admission/offloading decisions. |
| **Deadline / risk treatment** | Hard task deadlines are part of the feasibility checks; primary focus is long-term energy constraints, queue stability and latency/energy trade-off, not a contextual bandit or formal latency chance constraint. |
| **Baselines / metrics evidenced** | Local execution, random offloading, edge-only, and classical Lyapunov-based offloading; reports average delay and energy. |
| **Overlap with our project** | Adjacent prior art for adaptive online offloading under dynamic resources, queues and deadlines; algorithmic family differs from our contextual-bandit configuration. |
| **Defensible distinction / implication** | Dynamic offloading and queue-aware adaptation are not new alone. Our comparison should specify the learning policy and fair experimental protocol rather than claim general adaptive offloading novelty. |
| **Evidence grade** | Official SpringerLink full text reviewed. |
| **Publisher evidence** | Official SpringerLink full text |

### Risk-aware user satisfaction maximization in vehicle-assisted multi-access edge computing offloading: a game-theoretic approach

| | |
|---|---|
| **ID** | SD-08 |
| **Database** | ScienceDirect |
| **Year** | 2026 |
| **DOI** | 10.1016/j.dcan.2025.11.001 |
| **Link** | https://www.sciencedirect.com/science/article/pii/S2352864825001750 |
| **Methodology** | Prospect Theory models user risk-aware behavior; offloading is formulated as a non-cooperative game with a proven unique Pure Nash Equilibrium and a low-complexity distributed iterative optimization algorithm. |
| **Deadline / risk treatment** | Models uncertain vehicle-provided compute resources and risk-aware user satisfaction; abstract reports lower vehicle failure probability. Not a contextual bandit and not a formal per-task deadline chance constraint. |
| **Baselines / metrics evidenced** | Publisher abstract reports comparisons with benchmark methods; full baseline names and equations were not independently extracted in this pass. |
| **Overlap with our project** | Relevant prior art for risk-aware offloading under uncertain compute capacity, but the mechanism is game-theoretic and vehicle-assisted. |
| **Defensible distinction / implication** | Risk-aware task offloading is established broadly. Our current binary deadline penalty does not implement Prospect Theory or a calibrated risk model. |
| **Evidence grade** | Official ScienceDirect abstract and metadata reviewed; full equation/table extraction not completed. |
| **Publisher evidence** | Official ScienceDirect record |

# Core Papers

Supplied/core paper review matrix

Core literature comparison

8 supplied/relevant records counted once; the 9th record is an additional Elsevier publisher-page check, not a full PDF review. Author-stated limitations and our inferred limitations are explicitly separated.

## RL-Based Adaptive Task-Offloading in Mobile-Edge Computing for IoT Networks

| | |
|---|---|
| **ID** | P1 |
| **Venue/year** | IEEE Internet of Things Magazine, 2025 |
| **DOI / citation key** | DOI 10.1109/MIOT.2025.3623451 |
| **Dataset / workload evidence** | No public workload trace identified. Simulated LTE/NOMA ultra-dense MEC; approx. 10–100 active nodes, up to 10 tasks/node, 5 edge servers; model-generated task/network conditions. |
| **Simulator / testbed** | Simulation; modeled wireless MEC |
| **State / action / method** | Abstract/contribution says multi-agent DQN, but implementation section describes tabular Q-learning/Q-table; action relates to local vs edge/base-station assignment. |
| **Reward / objective** | Reward based on negative execution time; problem framing discusses energy and QoS delay constraints. |
| **Baselines** | Local, least-load / optimized variants (as described) |
| **Metrics / reported result scope** | Delay, energy, QoS/reliability, throughput-related outcomes. |
| **Explicit author limits / future work** | Explicit future directions: multi-objective latency/throughput/energy; joint computation/communication/caching; deep multi-agent RL; robustness/scaling. |
| **INFERRED limitations / open checks** | INFERRED: DQN vs Q-table inconsistency harms reproducibility; simplified base-station-to-edge delay; no identified public workload trace or held-out topology test. |

## Edge-Cloud Adaptive Framework for Real Time IoT Data Processing in Smart Cities

| | |
|---|---|
| **ID** | P2 |
| **Venue/year** | IEEE ACROSET, 2025 |
| **DOI / citation key** | DOI 10.1109/ACROSET66531.2025.11280634 |
| **Dataset / workload evidence** | 102 traffic-camera streams described as CityScape-Traffic replayed for 4 rush-hour hours; 64 air-quality sensors at 1 Hz; synthetic bursts triple camera arrivals for 120 s every 30 min. Public data link/licence unverified. |
| **Simulator / testbed** | Jetson Nano + AWS EKS; MQTT; WAN emulator |
| **State / action / method** | Double Dueling DQN; per-micro-batch Edge or Cloud; state includes cloud latency, backhaul bandwidth, CPU/GPU use, moving average latency. |
| **Reward / objective** | Normalized latency, bandwidth, energy (reward equation formatting in PDF is imperfect). |
| **Baselines** | Cloud-only; static fog |
| **Metrics / reported result scope** | Mean/P95 latency; 100-ms SLO satisfaction; backhaul; energy/frame; Jain fairness; overhead. |
| **Explicit author limits / future work** | Explicit future: policy-gradient alternatives, adaptive pruning; extra cloud GPU overhead for federated update is reported. |
| **INFERRED limitations / open checks** | INFERRED: few baselines; 4K in abstract vs 1920x1080 in dataset section; month-long training trace vs 4-hour replay requires clarification. |

## Deep Q-Network based Task Offloading Optimisation for Edge-IIoT System

| | |
|---|---|
| **ID** | P3 |
| **Venue/year** | IEEE ICICIT, 2026 |
| **DOI / citation key** | DOI 10.1109/ICICIT69063.2026.11634047 |
| **Dataset / workload evidence** | Mentions Kaggle/Edge-IIoT network data for telemetry/features. Separate fixed parameters include 100-KB task size, 500 mega-cycles, local CPU 1.5 GHz, edge 50 GHz, cloud 100 GHz, bandwidth 50 Mbps, transmit power 2 W. Network dataset ≠ complete scheduling benchmark. |
| **Simulator / testbed** | Simulation (implementation details need independent reproduction) |
| **State / action / method** | DQN actions Local/Edge/Cloud; features include task size/cycles/deadline, latency/energy/bandwidth/server load. |
| **Reward / objective** | Reward-based objective reports latency, energy, cost, deadline success and utilization; exact reproduction needs equation/implementation audit. |
| **Baselines** | Local-only, edge-only, cloud-only / offloading variants described in the paper. |
| **Metrics / reported result scope** | Latency, energy, cost, reward, utilization, on-time completion. |
| **Explicit author limits / future work** | No crisp formal limitations section found; general Edge-IIoT challenges discussed in paper. |
| **INFERRED limitations / open checks** | INFERRED: task workload generation/reward details under-specified; network telemetry and task CPU/deadline fields mixed conceptually; no held-out topology/confidence-interval evidence clear. |

## Deep Learning-Centric Task Offloading in IoT-Fog-Cloud Continuum: State-of-the-Art Review

| | |
|---|---|
| **ID** | P4 |
| **Venue/year** | IEEE Access, 2025 |
| **DOI / citation key** | DOI 10.1109/ACCESS.2025.3599190 |
| **Dataset / workload evidence** | Survey: no single dataset/workload applies to all reviewed papers. |
| **Simulator / testbed** | Survey, not a single simulator |
| **State / action / method** | No single state/action/reward. Provides taxonomy of DL/RL offloading approaches. |
| **Reward / objective** | Not applicable at survey level. |
| **Baselines** | Not applicable as one benchmark; paper surveys prior methods. |
| **Metrics / reported result scope** | Maps metrics, methods, simulator/testbed and future research. |
| **Explicit author limits / future work** | Survey discusses data scarcity, testbed/simulator limitations, energy, security, workload prediction and provisioning. |
| **INFERRED limitations / open checks** | INFERRED: survey-level gaps do not prove a specific method combination is unpublished; primary-paper comparison remains necessary. |

## An Adaptive Edge-Cloud Task Offloading Framework using RL for Latency-Sensitive AI Applications

| | |
|---|---|
| **ID** | P5 |
| **Venue/year** | IEEE ICOECA, 2026 |
| **DOI / citation key** | DOI 10.1109/ICOECA68095.2026.11485334 |
| **Dataset / workload evidence** | Parameterized/synthetic profiles: vision 8–20 MB, 3–8 Gcycles, 4–10 tasks/s; AR/VR 2–10 MB, 2–6 Gcycles, 6–14 tasks/s; health-IoT 0.5–3 MB, 1–4 Gcycles, 3–8 tasks/s; industrial CPS 1–6 MB, 2–7 Gcycles, 5–12 tasks/s. No public trace identified. |
| **Simulator / testbed** | Simulated edge/cloud framework |
| **State / action / method** | RL state includes edge utilization/queue, network rate, cloud queue and task features; action Edge/Cloud only. |
| **Reward / objective** | Penalty for latency plus deadline-met indicator. |
| **Baselines** | Threshold, greedy latency-minimization, Lyapunov/LODCO-like method. |
| **Metrics / reported result scope** | Mean/P95 latency, deadline success, queue/resource outcomes, energy (reported). |
| **Explicit author limits / future work** | Explicit future: trust modeling, privacy-conscious offloading. |
| **INFERRED limitations / open checks** | INFERRED: no device/local action, waiting or partial splitting; data source appears to be generated ranges, not public trace. |

## An Analysis of Task Offloading Approaches in Edge-Cloud Continuum

| | |
|---|---|
| **ID** | P6 |
| **Venue/year** | IEEE COMPSAC, 2025 |
| **DOI / citation key** | DOI 10.1109/COMPSAC65507.2025.00206 |
| **Dataset / workload evidence** | EdgeCloudSim sample_app4, configured/generated app profiles (AR, health, heavy compute, infotainment); 200–1000 devices moderate, 2000+ high load; 5 configurations, 10 iterations. No public task trace identified. |
| **Simulator / testbed** | EdgeCloudSim |
| **State / action / method** | Rule-based comparison, not learned policy; placement Local/Edge/Cloud depending on policy. |
| **Reward / objective** | Policy-specific heuristic objectives; not one common RL reward. |
| **Baselines** | NETWORK_BASED, UTILIZATION_BASED, FUZZY_BASED, FUZZY_COMPETITOR, HYBRID. |
| **Metrics / reported result scope** | Failure ratio, service time, placement distribution. |
| **Explicit author limits / future work** | Future work: combine fuzzy adaptability and network awareness. |
| **INFERRED limitations / open checks** | INFERRED: no DQN/RL comparator or external trace identified; results tied to configured sample_app4 workload. |

## Hierarchical DRL-Based Partial Task Offloading in Device-Edge-Cloud Architecture

| | |
|---|---|
| **ID** | P7 |
| **Venue/year** | IEEE Transactions on Consumer Electronics, 2026 |
| **DOI / citation key** | DOI 10.1109/TCE.2025.3611973 |
| **Dataset / workload evidence** | Synthetic tasks (data size, CPU cycles, tolerable latency sampled in given ranges); Network I/II; no public task trace identified. |
| **Simulator / testbed** | Authors' Python/PyTorch simulation; not stated to be iFogSim2/EdgeCloudSim |
| **State / action / method** | Hierarchical: high-level Double DQN chooses tier/actions; low-level PPO chooses split ratios across device, edge and cloud; fractions sum to one. |
| **Reward / objective** | Multi-objective energy, task acceptance and resource use subject to latency constraints. |
| **Baselines** | DRL-PTO (fixed all-tier split) and DRL-TO (whole-task placement). |
| **Metrics / reported result scope** | Reward; acceptance ratio; energy; generalization across 100 runs/new task profiles. |
| **Explicit author limits / future work** | Future: Transformer/quantum DRL, meta-learning, digital twins. |
| **INFERRED limitations / open checks** | INFERRED: only 2 network configurations, no external trace, limited classical/optimization baselines; two-agent complexity significantly higher. |

### Task Offloading for CAVs Edge Computing Environment: Taxonomy, Critical Review, and Future Road Map

| | |
|---|---|
| **ID** | P8 |
| **Venue/year** | ACM Computing Surveys, 2026 |
| **DOI / citation key** | DOI 10.1145/3783984 (ACM Computing Surveys 58(8), Article 194, Feb. 2026) |
| **Dataset / workload evidence** | CAV task-offloading survey; discusses KITTI, Waymo Open, BDD100K and ApolloScape as automotive perception datasets. These are not task CPU/deadline/network outcome traces aligned for an iFogSim offloading benchmark. |
| **Simulator / testbed** | Survey/taxonomy, not one experiment |
| **State / action / method** | Not applicable. |
| **Reward / objective** | Not applicable. |
| **Baselines** | Not applicable. |
| **Metrics / reported result scope** | Reviews workload, mobility, dataset, simulator and evaluation taxonomies. |
| **Explicit author limits / future work** | Discusses real-time reliability/latency, energy, mobility, collaboration, discovery and generalization. |
| **INFERRED limitations / open checks** | INFERRED: automotive mobility data should not be imported into stationary IoT study without an explicit motivation. |

### An efficient computational offloading method using deep reinforcement learning in edge-end-cloud

| | |
|---|---|
| **ID** | P9 |
| **Venue/year** | Ad Hoc Networks / Elsevier, 2025 |
| **DOI / citation key** | DOI 10.1016/j.adhoc.2025.103941 |
| **Dataset / workload evidence** | Full-text PDF not provided; dataset/workload not verified from page/abstract. |
| **Simulator / testbed** | Publisher abstract/page only; full text not audited. |
| **State / action / method** | Page describes threshold pre-stage, DQN + Q-learning, blockchain / integrity aspects in edge-end-cloud. |
| **Reward / objective** | Full reward/objective not verified from full text. |
| **Baselines** | Not fully verified. |
| **Metrics / reported result scope** | Publisher page says approximately 40% reduction against strategies; exact conditions and baselines not independently audited. |
| **Explicit author limits / future work** | Full-text limitations not verified. |
| **INFERRED limitations / open checks** | Classify as page/abstract-level evidence only; obtain PDF before making finer claims. |

## External Precedents

Adjacent prior-art scan — supporting evidence, not exhaustive

External overlap / novelty risk scan

Search cut-off 27 Oct 2026. These are close precedents, not an exhaustive proof of absence. Directly related papers should be checked in full text before manuscript submission.

### Optimization of Task Scheduling Strategy With Timing and Data Dependencies in VEC Based on DRL

| | |
|---|---|
| **Dimension** | Latest-start-time / timing-dependent scheduling |
| **Venue/year** | IEEE TMC, 2026 |
| **DOI** | 10.1109/TMC.2025.3646450 |
| **Relevant overlap found** | DAG timing/data dependencies; Latest Start Time metric; MADDPG |
| **Consequence for our claim** | Does not support novelty for ‘when to start’ alone. |
| **Evidence confidence / note** | IEEE Xplore search record / DOI verified |

## Joint optimization of multi-dimensional resource allocation and task offloading for QoE enhancement in Cloud-Edge-End collaboration

| | |
|---|---|
| **Dimension** | Partial offloading + resource allocation |
| **Venue/year** | Future Generation Computer Systems, 2024 |
| **DOI** | 10.1016/j.future.2024.01.025 |
| **Relevant overlap found** | Dependent tasks, resource allocation, latency/energy objective; MAPPO. |
| **Consequence for our claim** | Partial / multi-site task splitting is already studied. |
| **Evidence confidence / note** | ScienceDirect publisher record |

## Delay-aware partial task offloading using multicriteria decision model in IoT–fog–cloud networks

| | |
|---|---|
| **Dimension** | Partial offload + deadlines |
| **Venue/year** | JNCA, 2025 |
| **DOI** | 10.1016/j.jnca.2025.104278 |
| **Relevant overlap found** | Task partitioning, processing-efficiency criteria, deadlines and completion-time assignment. |
| **Consequence for our claim** | No claim that deadline-aware partial offload is new. |
| **Evidence confidence / note** | ScienceDirect page says no data were used for the study. |

## Forecast-driven task offloading for reliable and adaptive MEC

| | |
|---|---|
| **Dimension** | Forecast + contextual bandit |
| **Venue/year** | JNCA, 2026 |
| **DOI** | 10.1016/j.jnca.2026.104494 |
| **Relevant overlap found** | TimesNet forecast + LinUCB; real-world multivariate traces + simulated failures. |
| **Consequence for our claim** | Forecast + LinUCB is not a fresh novelty by itself. |
| **Evidence confidence / note** | ScienceDirect publisher page |

## Multi-armed bandit-based federated RL for dynamic task offloading in containerized MEC

| | |
|---|---|
| **Dimension** | Bandit + federated RL |
| **Venue/year** | Discover Applied Sciences, 2026 |
| **DOI** | 10.1007/s42452-026-08386-7 |
| **Relevant overlap found** | Device MAB + federated PPO/DRL at edge; Local/Edge/Cloud; deadlines. |
| **Consequence for our claim** | Bandit, hybrid MAB/RL, and three-tier placement already overlap. |
| **Evidence confidence / note** | Springer page; data-availability wording should be scrutinized. |

## LITO: Lemur-Inspired Task Offloading for Edge–Fog–Cloud Continuum Systems

| | |
|---|---|
| **Dimension** | Contextual bandit + deadline feasibility gate |
| **Venue/year** | Sensors, 2026 |
| **DOI** | 10.3390/s26051497 |
| **Relevant overlap found** | Contextual-bandit feedback + explicit capacity/deadline filtering and fallback. |
| **Consequence for our claim** | Very close precedent to naive bandit+deadline gate; exact algorithm and evaluation must be compared. |
| **Evidence confidence / note** | MDPI publisher page, adjacent publisher used for overlap check. |

## SLA-aware deep reinforcement learning for adaptive EdgeCloud task scheduling

| | |
|---|---|
| **Dimension** | Deadline/SLA-aware DRL |
| **Venue/year** | Scientific Reports, 2026 |
| **DOI** | 10.1038/s41598-026-40237-8 |
| **Relevant overlap found** | SLA/deadline risk used in DRL scheduling. |
| **Consequence for our claim** | Deadline-aware deep scheduling not new. |
| **Evidence confidence / note** | Nature publisher page |

## Guarded deep reinforcement learning for SLA latency and energy aware task scheduling in IoT edge cloud environments

| | |
|---|---|
| **Dimension** | Guarded DRL |
| **Venue/year** | Discover Computing, 2026 |
| **DOI** | 10.1007/s10791-026-10607-x |
| **Relevant overlap found** | Guarded DRL for SLA/latency/energy tradeoffs. |
| **Consequence for our claim** | Generic safe/guarded SLA scheduling has precedents. |
| **Evidence confidence / note** | Springer publisher page |

## Chance constraint-based dependent task offloading for MEC under uncertainty

| | |
|---|---|
| **Dimension** | Chance constraints + uncertainty |
| **Venue/year** | Applied Soft Computing, 2026 |
| **DOI** | 10.1016/j.asoc.2026.115710 |
| **Relevant overlap found** | Dependent task offloading and probabilistic constraints under uncertainty. |
| **Consequence for our claim** | Probabilistic deadline feasibility is not new in general. |
| **Evidence confidence / note** | ScienceDirect page |

## Dynamic task offloading strategy in MEC using meta-RL with adaptive learning rate

| | |
|---|---|
| **Dimension** | Meta-RL / adaptation |
| **Venue/year** | Journal of King Saud University C&IS, 2026 |
| **DOI** | 10.1007/s44443-026-00736-0 |
| **Relevant overlap found** | Synthetic DAGs; meta-RL/Seq2Seq; adaptation across task complexity. |
| **Consequence for our claim** | Generalization alone is not a novelty claim; specify shifts and transfer protocol. |
| **Evidence confidence / note** | Springer page |

## Generalizable Pareto-Optimal Offloading With RL in MEC

| | |
|---|---|
| **Dimension** | Generalizable Pareto offloading |
| **Venue/year** | IEEE TSC, 2025 |
| **DOI** | 10.1109/TSC.2025.3604371 |
| **Relevant overlap found** | Multiobjective RL, varying preferences and MEC configurations. |
| **Consequence for our claim** | Generic multiobjective generalization already exists. |
| **Evidence confidence / note** | IEEE Xplore record |

### Dynamic Task Offloading in Edge Computing Based on Dependency-Aware RL

| | |
|---|---|
| **Dimension** | Dependency-aware offloading + cluster trace |
| **Venue/year** | IEEE TCC, 2024 |
| **DOI** | 10.1109/TCC.2024.3381646 |
| **Relevant overlap found** | DAG/dependency-aware RL; Alibaba cluster trace plus synthetic workload. |
| **Consequence for our claim** | DAG scheduling and trace-based benchmarking already exist. |
| **Evidence confidence / note** | IEEE Xplore record |

### An efficient computational offloading method using DRL in edge-end-cloud

| | |
|---|---|
| **Dimension** | Edge-end-cloud DRL + thresholding |
| **Venue/year** | Ad Hoc Networks, 2025 |
| **DOI** | 10.1016/j.adhoc.2025.103941 |
| **Relevant overlap found** | Threshold pre-stage, DQN/Q-learning; page-level evidence only. |
| **Consequence for our claim** | Edge-end-cloud DRL plus thresholding already exists. |
| **Evidence confidence / note** | Full PDF was not reviewed; avoid detailed limitation claims. |

### Computation offloading in the edge-to-cloud compute continuum: a survey of federated architectural solutions

| | |
|---|---|
| **Dimension** | Workload and trace data gap |
| **Venue/year** | Cluster Computing, 2025 |
| **DOI** | 10.1007/s10586-025-05577-6 |
| **Relevant overlap found** | Survey notes lack of datasets that align task, device, deadline, energy and network dimensions. |
| **Consequence for our claim** | Supports disclosed synthetic benchmark, not a claim of dataset novelty. |
| **Evidence confidence / note** | Springer publisher page |

# Search Log

Combined search log — counts/exclusions still incomplete

### v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | IEEE/Institutional record |
| **Query** | "Chance-Constrained Task Offloading for Reliability Guarantees in Multi-Tenant Networks" |
| **Purpose** | Verify O2C/CCMAB method, venue, DOI |
| **Result / evidence** | Institutional record and DOI confirmed; abstract verified; full IEEE equations not extracted |
| **URL** | https://researchportal.ip-paris.fr/fr/publications/chance-constrained-task-offloading-for-reliability-guarantees-in-/ |

### v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | ScienceDirect |
| **Query** | "Forecast-driven task offloading for reliable and adaptive mobile edge computing" |
| **Purpose** | Verify recent forecast+bandit prior art |
| **Result / evidence** | JNCA 2026, TimesNet + LinUCB, load/failure forecasting; abstract verified |
| **URL** | https://doi.org/10.1016/j.jnca.2026.104494 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | MDPI Sensors full text |
| **Query** | "LITO" "heterogeneous deadlines" "contextual bandit" edge fog cloud |
| **Purpose** | Check overlap for deadline heterogeneity and node failure |
| **Result / evidence** | Full article reviewed; EDF-DRS, contextual bandit, feasibility filter, fallback, failure experiments are reported |
| **URL** | https://www.mdpi.com/1424-8220/26/5/1497 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | IIETA publisher full text |
| **Query** | "QoS-Aware Task Offloading" "Hybrid Deep Q-Learning" iFogSim2 |
| **Purpose** | Check same-simulator RL prior art |
| **Result / evidence** | H-DQAC 2025, hybrid DQL/Actor-Critic, QoS metrics, iFogSim2; full publisher page reviewed |
| **URL** | https://iieta.org/journals/jesa/paper/10.18280/jesa.581016 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | ScienceDirect |
| **Query** | "RELTO" reliability-oriented DRL context-aware adaptive reward weighting MEC |
| **Purpose** | Check reliability-aware reward overlap |
| **Result / evidence** | PPO hybrid actions; reliability/latency/energy objective; publisher abstract verified |
| **URL** | https://www.sciencedirect.com/science/article/pii/S1570870525003130 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | ScienceDirect |
| **Query** | "risk-regularised DNN-PPO" edge fog strict SLA CVaR |
| **Purpose** | Check tail-risk and SLA overlap |
| **Result / evidence** | CVaR regularization, SLA critic, p95/p99 evaluation; publisher page reviewed |
| **URL** | https://www.sciencedirect.com/science/article/pii/S0140366426002008 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | ScienceDirect |
| **Query** | "Chance constraint-based dependent task offloading" MEC uncertainty |
| **Purpose** | Check chance-constrained optimization overlap |
| **Result / evidence** | 2026 Applied Soft Computing; chance-constrained dependent-task offloading |
| **URL** | https://www.sciencedirect.com/science/article/pii/S1568494626011580 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | ScienceDirect / institutional record |
| **Query** | "Deadline-Aware joint optimization" cell-Free MEC 2026 |
| **Purpose** | Check deadline-aware RL and deadline-drop metrics |
| **Result / evidence** | Computer Networks 2026, DOI verified |
| **URL** | https://www.sciencedirect.com/science/article/pii/S1389128626002318 |

## v26 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | arXiv |
| **Query** | "Energy-Efficient Deadline-Aware Edge Computing" E2DA contextual neural bandit |
| **Purpose** | Verify earlier direct prior art |
| **Result / evidence** | Full preprint and IEEE GLOBECOM DOI metadata available |
| **URL** | https://arxiv.org/abs/2308.06647 |

## v29 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | SpringerLink |
| **Query** | Fuzzy chance constraint-based task offloading for MEC under uncertainties with joint multi-network collaboration and privacy preservation |
| **Purpose** | Official open-access full text: fuzzy chance constraints, multi-network allocation, privacy constraint, adaptive genetic algorithm + Monte Carlo + neural-network proxy; four baselines stated in article text. |
| **Result / evidence** | Adds direct evidence that uncertainty/chance-constrained offloading is prior art, though problem and method differ from our contextual policy. |
| **URL** | Need inspect the baseline comparison table before quoting baseline names. |

## v29 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | SpringerLink |
| **Query** | Lyapunov Optimization with Virtual Queues for adaptive task offloading in multi access edge computing |
| **Purpose** | Official open-access full text: Lyapunov Drift-Plus-Penalty, virtual energy queues, decay-aware updates, battery-aware admission, local/edge decisions, four baselines. |
| **Result / evidence** | Adjacent prior art for online queue-aware energy-delay offloading; supports avoiding broad claims about adaptive offloading. |
| **URL** | Different algorithm family; not a contextual-bandit paper. |

## v29 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | ScienceDirect |
| **Query** | Risk-aware user satisfaction maximization in vehicle-assisted multi-access edge computing offloading: a game-theoretic approach |
| **Purpose** | Official publisher abstract/metadata: Prospect Theory risk-aware behavior, non-cooperative game, unique Pure Nash Equilibrium, uncertain vehicle compute resources. |
| **Result / evidence** | Adds prior-art evidence for risk-aware offloading under uncertain capacity. |
| **URL** | Full equations and baseline names were not extracted in this pass. |

## v29 targeted search

| | |
|---|---|
| **Date** | 2026-10-28 |
| **Database/source** | Count / quality correction |
| **Query** | Audit of v28 Literature Matrix and report statements |
| **Purpose** | The v28 matrix contains 18 records: 14 additions (7 ScienceDirect, 4 SpringerLink, 3 IEEE records) plus 4 carried-forward anchors. |
| **Result / evidence** | Corrected prior report statement that said 13 additions plus 5 anchors; v29 matrix contains 21 records after adding 3 newly reviewed papers. |
| **URL** | The database search remains targeted, not exhaustive native-interface export. |

# Action Plan

Execution plan and closure gates

## Supervisor decision

| | |
|---|---|
| **Priority** | P0 |
| **Specific actions** | Choose comparative/reproducibility study vs genuinely new algorithm. Confirm primary simulator and expected contribution level. |
| **Owner** | Supervisor + Mark + team |
| **Deliverable/evidence** | Written decision and agreed research question |
| **Exit criterion** | No new method built until this is confirmed. |
| **Dependency/status** | Immediate / open |

## Version/provenance

| | |
|---|---|
| **Priority** | P0 |
| **Specific actions** | Tag/freeze v26 code, model artifact, config, topology, training seeds and held-out seeds; hash the release and preserve raw JSONL. |
| **Owner** | Simulator owner + Mark |
| **Deliverable/evidence** | Version sheet + manifest + code/model/config identifiers |
| **Exit criterion** | Every result row traces to one exact configuration. |
| **Dependency/status** | Immediate / v26 main; v25 and Seed1/2 separate |

## Algorithm naming

| | |
|---|---|
| **Priority** | P0 |
| **Specific actions** | If γ=0 / contextual_bandit config remains, call it tabular contextual-bandit/Q-value learner. If sequential Q-learning is desired, implement γ>0 and test delayed consequences. |
| **Owner** | Model owner + supervisor |
| **Deliverable/evidence** | Algorithm description matching config and code |
| **Exit criterion** | Paper, report and code use consistent name. |
| **Dependency/status** | Immediate |

## Native database evidence

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Export searches from IEEE Xplore, SpringerLink and ScienceDirect; save query, date, result count, export and screening columns. Record duplicate handling and exclusion reason. |
| **Owner** | Research Gap owner |
| **Deliverable/evidence** | Native exports + screening log |
| **Exit criterion** | Search trail is reproducible; no invented result counts. |
| **Dependency/status** | After meeting; focused v29 audit done |

## Full-text equation audit

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Prioritize E2DA, O2C/CCMAB, LITO, H-DQAC, forecast-driven LinUCB, RELTO, CVaR and other closest papers. Mark equation/baseline inaccessible if paywalled. |
| **Owner** | Research Gap owner + Member 1 |
| **Deliverable/evidence** | Claim-by-claim equation and baseline matrix |
| **Exit criterion** | Each novelty distinction linked to text/equation or labelled uncertain. |
| **Dependency/status** | Some source evidence only abstract-level |

## Task-level logging

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Capture task_id, arrival, decision, action, chosen site, transfer start/end per hop, queue start/end, actual CPU start/end, first-module finish, full-loop finish, deadline, status/reason. |
| **Owner** | Simulator owner |
| **Deliverable/evidence** | Schema + validated raw logs |
| **Exit criterion** | One terminal status per task; no mismatches/orphans. |
| **Dependency/status** | Needs code instrumentation |

## Metric definitions

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Define deadline relative to arrival; absolute deadline=arrival+relative deadline; fix late/failed/pending/censored denominator rules; never drop pending silently. |
| **Owner** | Simulator owner + supervisor |
| **Deliverable/evidence** | Metric specification sheet |
| **Exit criterion** | Every metric reproducible from logs; units validated. |
| **Dependency/status** | Before formal experiments |

## Replace proxies

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Implement actual CPU utilization and per-hop bytes/transfer times. Add energy only if model, units and accounting are validated. |
| **Owner** | Simulator owner |
| **Deliverable/evidence** | Telemetry and unit tests |
| **Exit criterion** | No CPU/network/energy claim based solely on proxy. |
| **Dependency/status** | Before resource/energy claims |

## Baseline cleaning

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Run Local/Edge/Cloud/Random/latency-greedy/current policy; remove deadline-greedy duplicate or redesign as truly distinct. Add DQN only if justified. |
| **Owner** | Algorithm owner |
| **Deliverable/evidence** | Pinned baseline implementations and configs |
| **Exit criterion** | Same workload/seed, same simulator and metric definitions. |
| **Dependency/status** | Before confirmatory runs |

## Confirmatory stats

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Freeze policies and tuning; run ≥20 paired seeds (target 30 if feasible), same task generation within seed, fresh held-out test seeds, report mean/P95/P99/CI and task-type strata. |
| **Owner** | Whole team |
| **Deliverable/evidence** | Final result CSV + test protocol |
| **Exit criterion** | No tuning on final test; report variability and confidence intervals. |
| **Dependency/status** | After metric and policy freeze |

### Generalization

| | |
|---|---|
| **Priority** | P1 |
| **Specific actions** | Evaluate at least 2–3 topologies or carefully justified workload/network trace proxies; test load, WAN delay/bandwidth, edge-capacity and outage shifts separately first. |
| **Owner** | Whole team |
| **Deliverable/evidence** | Scenario matrix + manifests |
| **Exit criterion** | Repeatable tests with isolated factors and explicit limits. |
| **Dependency/status** | After base correctness |

### New risk method if required

| | |
|---|---|
| **Priority** | P2 |
| **Specific actions** | Specify residual quantile or other risk-calibration mechanism mathematically; distinguish it from LITO feasibility filtering and O2C chance constraints; tune only on validation. |
| **Owner** | Algorithm owner + research owner |
| **Deliverable/evidence** | Algorithm equations + ablation + baseline protocol |
| **Exit criterion** | Evidence of substantive difference and held-out benefit, else reject novelty claim. |
| **Dependency/status** | Only if supervisor requires a new method |

### Dataset realism

| | |
|---|---|
| **Priority** | P2 |
| **Specific actions** | Use documented synthetic generator as primary; optional Alibaba Cluster Trace for workload-shape proxy, Edge-IIoTset for network telemetry proxy, Topo4MEC for topology only. |
| **Owner** | Data owner |
| **Deliverable/evidence** | Generator/config/data mapping and license notes |
| **Exit criterion** | No source is mislabelled as complete task-offloading data. |
| **Dependency/status** | After primary benchmark is sound |

## Practical Requirements

Practical requirements extracted from papers + project audit

### Per-task task model

| | |
|---|---|
| **Derived from literature / experiments** | Deadline-aware / QoS papers; current v26 scenario |
| **Current state** | MI/data class present; v26 uses 20/50/120 deadline profile |
| **Required action** | Log task arrival, MI, input bytes, class, relative+absolute deadline and seed/profile. |
| **Acceptance criterion** | Deadline does not depend on selected site/policy; one task ledger row per generated task. |

### Context state

| | |
|---|---|
| **Derived from literature / experiments** | DQN/Bandit/forecast-driven papers |
| **Current state** | MI, WAN latency, backlog estimate; utilization remains proxy |
| **Required action** | Verify feature units/ranges and add queue/resource signals only if measured. |
| **Acceptance criterion** | State values reconstructable from logged telemetry; no train-test leakage. |

### Action space

| | |
|---|---|
| **Derived from literature / experiments** | Local/Edge/Cloud papers |
| **Current state** | Three actions; Edge target may be parent edge not separate Edge-1/2 action |
| **Required action** | Document action mask and actual target; decide if edge nodes should be separate actions. |
| **Acceptance criterion** | Action in policy log matches executed site; zero routing mismatch. |

## Objective / reward

| | |
|---|---|
| **Derived from literature / experiments** | E2DA, O2C, LITO, H-DQAC, RELTO, CVaR studies |
| **Current state** | γ=0 contextual-bandit config; reward/latency/deadline flags need frozen definition |
| **Required action** | State objective explicitly; no chance guarantee unless modelled. |
| **Acceptance criterion** | Reward/constraints mathematically specified, unit tested, and not changed between baselines. |

## Baselines

| | |
|---|---|
| **Derived from literature / experiments** | Static baselines and strong deterministic heuristics in supplied works |
| **Current state** | Local/Edge/Cloud/latency-greedy; Q policy; duplicate deadline-greedy; random in other configs |
| **Required action** | Freeze baseline roster and code; avoid counting duplicate implementations separately. |
| **Acceptance criterion** | Same scenario, seed, workload, cutoff, drain and metrics for every policy. |

## Latency/Deadline metrics

| | |
|---|---|
| **Derived from literature / experiments** | Vashisth, Swapna, H-DQAC; current output |
| **Current state** | Sensor-to-first-module completion; deadline violations and P95/P99 available |
| **Required action** | Add full-loop completion and per-task outcome; ensure all units consistent. |
| **Acceptance criterion** | Metric names match exactly what is measured; pending/censored is explicit. |

## Resource and network accounting

| | |
|---|---|
| **Derived from literature / experiments** | Papers using throughput, bandwidth and utilization |
| **Current state** | Backlog proxy, payload bytes proxy; per-hop total not complete |
| **Required action** | Instrument CPU start/finish, queue time, link transfer/bytes on each hop. |
| **Acceptance criterion** | No utilization or total-traffic claim based on proxy. |

## Failure semantics

| | |
|---|---|
| **Derived from literature / experiments** | LITO, forecast-driven MEC, outage test |
| **Current state** | Edge-1 service masking for new assignments; router active/in-flight tasks continue |
| **Required action** | Keep compute-service outage distinct from host crash/network failure. |
| **Acceptance criterion** | Scenario definition includes what happens to in-flight tasks and recovery. |

## Robustness / uncertainty

| | |
|---|---|
| **Derived from literature / experiments** | O2C, forecast-LinUCB, CVaR and workload-shift studies |
| **Current state** | Several synthetic scenarios; v26 only five test seeds per scenario-policy and one topology |
| **Required action** | Use paired fresh seeds, 20 minimum / target 30, 2–3 topologies if feasible. |
| **Acceptance criterion** | Report mean and CI, tails, misses and stratified results. |

## Data provenance

| | |
|---|---|
| **Derived from literature / experiments** | Continuum survey and data audit |
| **Current state** | No fully aligned public dataset verified |
| **Required action** | Document synthetic generator; optional partial proxies separately. |
| **Acceptance criterion** | Every task input is traceable to generator/trace mapping, version, seed and assumptions. |

## Reproducibility

| | |
|---|---|
| **Derived from literature / experiments** | All empirical comparisons |
| **Current state** | Code and raw logs bundled; hashes/commit and native-search exports still need final consolidation |
| **Required action** | Store model, config, seed list, run manifest, raw logs, code release ID. |
| **Acceptance criterion** | Independent team member can reproduce one seed and regenerate summary. |

## Research-gap evidence

| | |
|---|---|
| **Derived from literature / experiments** | 21-row targeted source matrix |
| **Current state** | No broad algorithmic gap confirmed; evidence grade varies |
| **Required action** | Complete publisher records + equation-level check; final search exports and exclusions. |
| **Acceptance criterion** | Final claim is narrow, sourced and approved by supervisor. |

# Data Options

Dataset/workload option audit

Dataset/workload source audit

No single public dataset verified as containing all required task, deadline, network, resource and outcome fields in one aligned record. Public sources can inform one component at a time.

## Alibaba Cluster Trace Program, v2017/v2018

| | |
|---|---|
| **What it actually contains** | Real production cluster traces; v2017 ~1300 machines / 12 h; v2018 ~4000 machines / 8 days and task DAGs. |
| **Useful role** | Workload / DAG-arrival-shape proxy, not full IoT edge data. |
| **What it does not provide** | Does not inherently provide per-task relative deadlines, network delays, IoT input sizes and iFogSim2 execution outcomes aligned in one record. |
| **License / usability checks** | Data downloads may require a brief survey; use research/study purposes and follow the trace docs. Check dataset version, citations and policy before using. |
| **Source URL** | https://github.com/alibaba/clusterdata |

## Edge-IIoTset

| | |
|---|---|
| **What it actually contains** | Public IIoT/cybersecurity/network telemetry dataset; source paper DOI 10.1109/ACCESS.2022.3165809. |
| **Useful role** | Optional network-event / telemetry proxy only, with clear mapping. |
| **What it does not provide** | Not a native task-offloading benchmark: cannot assume CPU cycles, per-task deadlines or site-specific execution outcomes come from it. |
| **License / usability checks** | Review dataset terms and license on the official page; do not relabel network records as compute jobs. |
| **Source URL** | http://ieee-dataport.org/8939 |

## Topo4MEC

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|---|---|
| **What it actually contains** | Topology edge-list files and ingress nodes; includes random graphs and Milan City Center topology derived from OpenCellID. |
| **Useful role** | Optional topology stress/generalization after core benchmark works. |
| **What it does not provide** | Topology only, not task-arrival/deadline/compute workload. |
| **License / usability checks** | Repository README states GPL-3.0; verify license/attribution requirements and whether its abstraction fits the current iFogSim topology. |
| **Source URL** | https://github.com/bnxng/Topo4MEC |

## iFogSim2

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|---|---|
| **What it actually contains** | Simulator toolkit; repository states mobility support, migration, microservice orchestration and dynamic distributed clustering. |
| **Useful role** | Execution/simulation platform. |
| **What it does not provide** | Not a dataset and doesn't remove need to verify event timing and resource accounting. |
| **License / usability checks** | Pin the git commit and configuration for reproducibility. |
| **Source URL** | https://github.com/Cloudslab/iFogSim |

## Cloud-Edge Collaborative Task Scheduling GitHub repo

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|---|---|
| **What it actually contains** | README describes Topo4MEC/Pakistan topology use and synthetic task attributes/deadlines; no license verified in scan. |
| **Useful role** | Possible inspiration / audit candidate only. |
| **What it does not provide** | Do not rely on it until license, data schema, generator, commit and runtime are checked. |
| **License / usability checks** | If adopted, write a small independent validation and preserve attribution; not a validated primary dataset yet. |
| **Source URL** | https://github.com/henrenzhenderen-wangyu/Cloud-Edge-Collaborative-Task-Scheduling |

## Main generated workload (recommended)

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|---|---|
| **What it actually contains** | Generate arrivals, bytes, MI/cycles, class and relative deadline from a documented configuration; at least steady Poisson, bursty/on-off, mixed task/deadline profiles. |
| **Useful role** | Primary benchmark because it can be aligned exactly with iFogSim2 task/event logging. |
| **What it does not provide** | Synthetic, therefore no claim of real-world representativeness without calibration and sensitivity analysis. |
| **License / usability checks** | Use fixed seeds, publish the generator/configuration; choose task deadlines independently of the chosen policy/site; include ranges from the reviewed papers as scenario settings, not as validated ground truth. |
| **Source URL** | Proposed, not an external dataset. |